

C-MVH-Min*: Compact Multi-Valued Heuristic Search under Min Ordering

One-page idea summary — bi-objective case

1. The bottleneck

MVH search (L-NAMOA_{dr}*, MVH, Wolff et al.) is locked to **Lex** ordering, because dimensionality reduction (DR) and *t*-discarding require it. Lex forces every node to be *bound* to one concrete surviving heuristic: $h = \min_{\text{lex}} \{h \in H(s) \mid \neg \exists t \in T : t \preceq \text{Tr}(g + h)\}$. Returning a *specific element* of a *g*- and *T*-dependent subset is a **search**, so computing a node's priority costs $\Theta(|H(s)|)$ — *for every path reaching s*. Ordering and filtering are inseparable.

2. The key observation

Let $h_{\text{id}}(s) = (\min_h h_1, \min_h h_2)$ be the *ideal point* (one per state, $O(1)$), and let $\mu(f) = \min(\bar{f}_1, \bar{f}_2)$ be the primary **Min** value. Then

$$\min_{h \in H(s)} \mu(g + h) = \mu(g + h_{\text{id}}(s))$$

an **exact identity, not a relaxation**: min distributes over a componentwise minimum, and it survives any coordinatewise-increasing normalization.

- **Fails for Max**: $H = \{(0, 10), (10, 0)\}$ gives 10 vs. 0 — gap unbounded.
- **For Lex** it holds only through a *member* of $H(s)$, which the filter must first find.

⇒ **Under Min, ordering decouples from filtering**. Changing the OF *removes* the scan; it does not merely accelerate it.

3. Consequences

A node is just (s, g) , implicitly standing for the whole set $F(s, g) = \text{ND}\{g + h\}$. Its key is an **SVH** evaluation, so Skyler et al.'s **Min constant-time dominance checks**, their correctness theorem, monotone extraction and the **Done** early-termination flag all transfer to the MVH setting *unchanged*. Two consistent layers — we sort on exactly what we prune on:

Layer	Driven by	Gives
Ordering + PDC/SDC	$h_{\text{id}}(s)$ (admissible SVH: $h_{\text{id}} \preceq h \preceq c(\pi)$)	$O(1)$ CTDC, <i>Done</i>
Extra pruning	full 2-D MVH $H(s)$	strictly stronger

4. The MVH layer is provably not redundant

Query the *existential* oracle $\text{SURVIVES}(s, g, T) = [\exists h \in H(s) : \neg(T \preceq g + h)]$ over a corner-annotated index of $H(s)$ (lo/hi per subtree, **both** objectives, no truncation):

- **Test A** (reject, $O(1)$, *free*): $T \preceq g + h_{\text{id}} \Rightarrow$ every $g + h$ dominated. Free because $g + h_{\text{id}}$ **is** the node's key — the ordering key doubles as the root pruning aggregate.
- **Test B** (accept, $O(1)$): the two staircase extremes; a stored *member* is a genuine witness. (No accept rule on hi — the upper corner is *more* likely dominated, so it certifies nothing.)
- Else $O(\log |H(s)|)$ descent, pruning any subtree whose lo is dominated.

One-sidedness is the point: $H(s) = \{(1, 8), (3, 5), (4, 4), (6, 2), (9, 1)\}$, $g = (6, 6)$, $T = \{(8, 5), (5, 9)\}$. Then $g + h_{\text{id}} = (7, 7)$ *survives*, so no single-valued reasoning can prune — yet all five real evaluations $(7, 14), (9, 11), (10, 10), (12, 8), (15, 7)$ are dominated. The MVH prunes; an SVH cannot.

5. Added value

vs.	What C-MVH-MIN* adds
Wolff (Maya)	First non-lex ordering for MVH. Key: $\Theta(H(s)) \rightarrow O(1)$ <i>exact</i> . <i>t</i> -discarding $\rightarrow O(1)$ CTDC. Survival scan $\rightarrow O(1)/O(\log)$. No DR, no ordering condition, no untruncated fallback, no pop-then-reinsert. Early termination.
Skyler (Shawn)	First extension of Min + CTDC from SVH to MVH , with strictly stronger pruning and one node per physical path.
Roi	He keeps Lex + DR and CHOOSEH's selection semantics, so he indexes <i>truncated</i> $(d-1)$ -dim vectors and degenerates in BOS. Dropping DR keeps $H(s)$ fully 2-D.
Shahaf	<i>Complementary, different indexed object</i> . He indexes the path frontier $F(v)$ (point-vs-set, DR-truncated, collapses to a scalar at $M=2$). We index the heuristic set $H(s)$ (set-vs-set, full 2-D). His $M=2$ $O(1)$ result is a property of DR on $F(v)$ and does not transfer. He explicitly lists <i>multi-valued heuristics</i> as an open question.

6. Honest caveats

- Only the *primary* Min value is exact; the tie-break component is optimistic.
- h_{id} is weaker per-node guidance than Maya's bound h — but under Min it is *forced*, being the exact key of the node's whole set. Net effect is an **empirical** question.
- SURVIVES is $O(|H(s)|)$ in the worst case; the sorted-staircase merge is the honest baseline.
- Normalization is mandatory for Min. *Done* gave Skyler et al. only ~3%.
- **Open problem**: tightening a node's key after extraction breaks the CTDC proof — the live set depends on g , so h_{id} is no longer a fixed SVH and monotone extraction fails.

7. Beyond $d = 2$

Lemma 1 holds verbatim in any dimension. Two blockers: (i) CTDC-Min for $d > 2$ is open in Skyler et al.; (ii) the oracle becomes genuinely high-dimensional, where by Shahaf's separation result no total order is sublinear — **this is where a geometric index finally earns its keep**, suggesting the natural combination: his index over $F(v)$, ours over $H(s)$.